

Secure Ubuntu Web Server with IDS/IPS

Project Overview

This project demonstrates the deployment and hardening of a secure Ubuntu Server environment using multiple layers of security controls within VMware Workstation.

Technologies Used

- Ubuntu Server 26.04 LTS
- Apache2 Web Server
- OpenSSH Server
- UFW Firewall
- Fail2Ban
- Suricata IDS
- VMware Workstation

Security Features Implemented

- Apache2 web server deployment and testing
- Secure remote administration via SSH
- Firewall protection using UFW
- Brute-force attack mitigation with Fail2Ban
- Network traffic monitoring using Suricata IDS
- Security monitoring and validation

Skills Demonstrated

Linux Administration, Web Server Deployment, Network Security, Firewall Configuration, Intrusion Detection, Intrusion Prevention, Security Monitoring, Virtualization, and Troubleshooting.

Project Outcome

The project successfully deployed a functional and secured Ubuntu web server environment. Multiple security layers were implemented and validated, providing practical experience with server hardening and intrusion detection technologies.